```
define ETM_CLOCKGATE
define CORE_CLOCKGATE
```

In r2p0, it is controlled by the single CLKGATE_PRESENT parameter for each instantiation of Cortex-M3, as described in Cortex-M3 integration and implementation manual available to RTL licensees. Control of clock gating is inferred from circuit activity. To minimise power consumption, ARM recommends the use of both architectural clock gating and leaf cell clock gating, inferred by synthesis tools for ASIC (chip) implementation by RTL licensees.

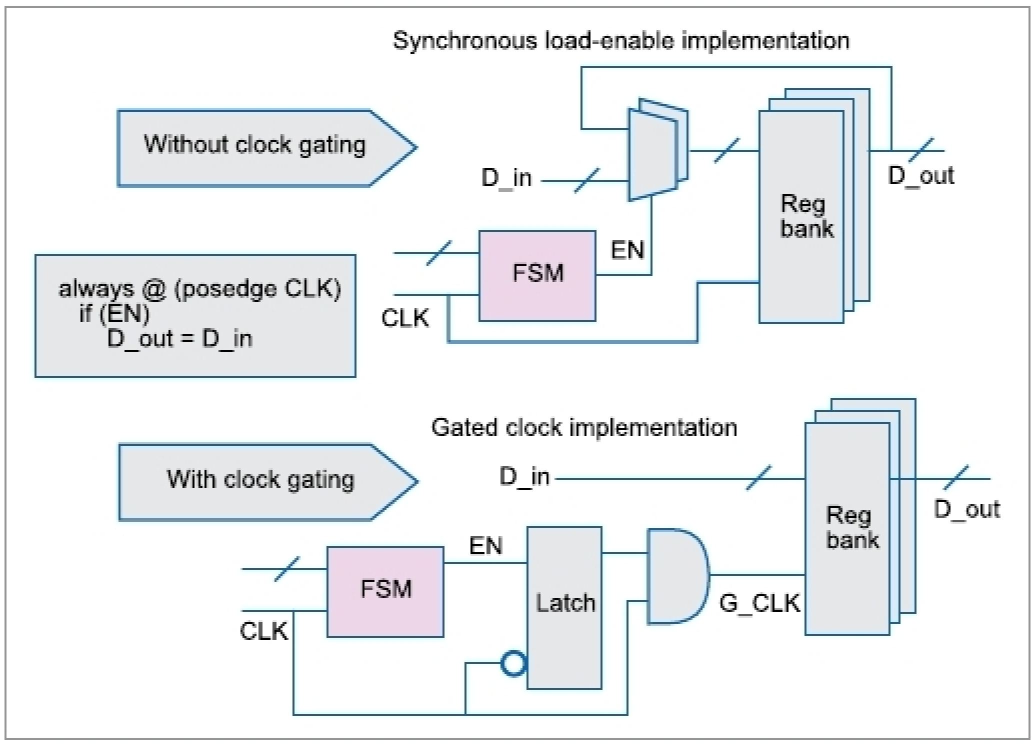

Fig. 2: Typical multi-bit flip-flop logic with enable

For FPGA prototyping, clock gating can cause some difficulty and inconvenience. However, as the benefit is relatively small in FPGA prototypes, it makes sense to de-configure clock gating in this case.

Architectural clock gating is a context-aware technique to determine functional idle period to optimise power. While designing the architecture, designers can identify the blocks for which clocks can be turned off under certain conditions. These conditions must be auto-detected by RTL or programmed through the software. Then, clocks can be turned off for that block, thereby reducing significant amount of dynamic power.

Architectural clock gating is mainly based on foreseeing which part of the design will be inactive under what condition, and gate that part of the design accordingly. It can evolve in two ways: clock gating based on protocol/design of IP during implementation, and analyse design power after implementation and optimise power using clock gating.

Programmable clock gating is done via a user programmable control register that determines the entry and exit of clock gating. This method is useful if exit and entry durations are high (considering software latency).

A disadvantage to this method is that, while disabling clock gating, if software latency is too high, it may lead to functional errors. A modification to the method is to use software for entry and exit using additional wakeup logic. Further modification is to provide a control register bit to enable clock gating for safety purposes, and logic for entry and exit.

Adaptive clock gating can be applied if complete IP is modelled into a single finite state machine (FSM) with several states. These can be divided into working and idle states. In this method, designers can gate the clock to complete IP during idle state.

There are no architecturally-defined FSMs. This method is used for a floating point ALU, which has several pipelines. When there are no instructions in the bus, it is used for gating the clock, assuming it is in idle state of IP.

In most IPs based on standard protocols, protocols themselves define the idle states for low-power mode. This method is helpful for IPs having single clock domain. These days, most IPs have multiple clock domains, which have scenarios where one clock domain is inactive, while the other is active. In those cases, it is not easy to gate the clock using this method.

Dynamic frequency variation

Dynamic frequency variation is a technique in which frequency of a particular block is dynamically increased to perform some operation. Once the operation is finished, clock frequency is decreased to original low frequency. For example, I2C controller application interface normally operates at lower frequency, but during data transfer, application interface clock frequency can be dynamically increased to finish data transfer quickly. Thus, average dynamic power consumption is reduced. The system can be made low power by considering low-power RTL design, gate-level optimisation, frequency islands, power gating, multi-supply voltage, multiple threshold voltage, dynamic voltage scaling and so on.

Most IoT applications use existing protocols such as USB and I2C, and ultra-low power fabrication processes. Given that most IoT devices are battery-powered, small power budgets are a strict design requirement. To this end, designers must take care of implementing low-power strategies at every level of abstraction, including inter-chip communication, SoC micro-architecture and technology node.

IP vendors are also updating their designs to meet the specific needs of the IoT market. Some popular low-power approaches that have been in use for some time are: clock gating, gate-level power optimisation, multi-VDD, multi-VT, power gating and adaptive voltage scaling. Reducing dynamic power is a major design focus for specialised IoT-optimised ultra-low power fabrication processes.

Clock gating can reduce dynamic power consumption applied at both front-end and back-end design phases. It can broadly be implemented at register or architectural level. With recent advancements in EDA tools, clock gating at register level is handled more efficiently. Clock gating is deployed at architecture level based on specific conditions unique to design and usage.